

UNISOLE SKILL AI LABS

Scientific AI & Machine Learning Professional Program

(Mathematics • Machine Learning • Deep Learning • Scientific Computing • PINN)

Duration: 6 Months

Mode: Live Online

Classes: Saturday + Sunday

Class Duration: 2 hours/session

Total Live Instruction: ~104 hours

Recommended Self-Learning: 4–6 hours/week

Level: Undergraduate → Early Professional

Project: Industry/Research Capstone

Core philosophy:

**Mathematics → Python → Scientific Computing → Machine Learning → Deep Learning
→ Scientific AI → Capstone**

1. Prerequisites

Eligible students

- B.Sc.: Physics / Mathematics / Statistics / Computer Science / Data Science
- B.Tech / B.E.
- B.Tech students from science/engineering branches
- M.Sc. students
- Students from Mathematics, Statistics, Physics, Engineering or other quantitative disciplines; Working professionals with basic programming/mathematics background

Required before starting Students should have:

- Basic school-level mathematics
- Basic calculus
- Basic algebra
- Basic computer literacy

Python programming is NOT mandatory

PROGRAM STRUCTURE		
Module	Duration	Main Focus
Module 1: Mathematical & Computational Foundations for Scientific AI	Weeks 1–4 (4 Weeks)	Python, Linear Algebra, Calculus, Probability, Gradient-Based Methods & Computational Foundations
Module 2: Scientific Computing, Numerical Methods & Differential Equations	Weeks 5–8 (4 Weeks)	NumPy, SciPy, Numerical Methods, ODE/PDE Solvers & Scientific Simulation
Module 3: Machine Learning for Scientific & Engineering Applications	Weeks 9–12 (4 Weeks)	Regression, Classification, Clustering, Model Evaluation, Feature Engineering & Scientific ML Baselines
Module 4: Deep Learning, PyTorch & Differentiable Computing	Weeks 13–16 (4 Weeks)	Neural Networks, Backpropagation, PyTorch, Automatic Differentiation & Deep Learning Projects
Module 5: Scientific Machine Learning & Physics-Informed Neural Networks (PINNs)	Weeks 17–21 (5 Weeks)	Scientific ML, Physics-Based Modelling, PINNs, Forward/Inverse Problems, PDE Constraints & Numerical Validation
Module 6: Advanced Scientific AI, Research Engineering & Industry Capstone	Weeks 22–26 (5 Weeks)	Advanced PINNs, Neural Operators, Research Computing, Benchmarking, Reproducibility, GitHub Portfolio & Capstone Defence
Total Duration	26 Weeks (~6 Months)	4 Contact Hours/Week · 104 Live Instruction Hours

DETAILED CURRICULUM BREAKDOWN

MODULE 1 — MATHEMATICAL & COMPUTATIONAL FOUNDATIONS FOR AI

Module Name: Mathematical & Computational Foundations for Scientific AI Module Code: UN-SAI-101 Module Type: Core		
Contact Hours/Week: 3L + 1T Module Credits: 04		
Module Objectives: Develop the mathematical, programming, and computational foundations required to implement scientific machine learning models. Students will translate mathematical formulations into working Python code and establish reproducible computational workflows.		
Units	Topics & Concepts	Project-Based Learning / Practical
Week 1	Scientific AI landscape; mathematical modelling; Python fundamentals; functions, loops, data structures	Set up Python, Jupyter/Colab and GitHub. Build a Python-based scientific calculator and document it.
Week 2	Linear algebra for AI; vectors, matrices, norms, matrix operations, eigenvalues, eigenvectors	Implement vector/matrix operations and visualize transformations. Mini-project: linear system solver.
Week 3	Calculus for AI; derivatives, partial derivatives, gradients, chain rule, Jacobian and Hessian intuition	Implement numerical differentiation and gradient descent. Compare numerical and analytical gradients.
Week 4	Probability, statistics, uncertainty, numerical error, loss functions and reproducible experiments	Mini-project: simulate a noisy scientific dataset, fit a mathematical model, quantify error, and publish a reproducible notebook.
Course Outcomes - CO1: Translate mathematical concepts into computational implementations. - CO2: Implement basic numerical and gradient-based methods. - CO3: Build reproducible Python notebooks and maintain code using Git/GitHub.		
Books & References		

1. Mathematics for Machine Learning — Deisenroth, Faisal & Ong
2. Python Data Science Handbook — Jake VanderPlas
3. Essential Mathematics for Data Science — Thomas Nield

MODULE 2: SCIENTIFIC COMPUTING, NUMERICAL METHODS & DIFFERENTIAL EQUATIONS

Module Name: Scientific Computing, Numerical Methods & Differential Equations Module Code: UN-SAI-102 Module Type: Core		
Contact Hours/Week: 3L + 1T Module Credits: 04		
Module Objectives: Enable students to solve scientific and engineering problems numerically, understand the relationship between differential equations and computational models, and generate reliable simulation datasets for ML and PINN training.		
Units	Topics & Concepts	Project-Based Learning / Practical
Week 5	NumPy, SciPy, Pandas; vectorization, data handling, scientific data pipelines	Develop a scientific data preprocessing pipeline with validation and visualization.
Week 6	Numerical errors, stability, convergence, root finding, interpolation and numerical differentiation	Compare numerical methods on a mathematical problem; analyze error and convergence.
Week 7	Numerical integration; ODEs; Euler, Runge–Kutta methods; initial-value problems	Build a numerical simulator for a harmonic oscillator or other dynamical system.
Week 8	PDE fundamentals; diffusion, heat and Poisson equations; boundary/initial conditions; finite-difference intuition	Implement a 1D heat-equation solver. Generate reference data and compare numerical solutions under different grid resolutions.
Course Outcomes		

1. CO1: Implement numerical algorithms for scientific problems. 2. CO2: Solve and visualize basic ODE/PDE systems numerically. 3. CO3: Generate, validate and interpret numerical simulation datasets.
Books & References 1. Scientific Computing: An Introductory Survey — Michael T. Heath 2. Numerical Methods for Engineers — Chapra & Canale 3. Numerical Recipes — Press et al.

MODULE 3: MACHINE LEARNING FOR SCIENTIFIC & ENGINEERING APPLICATIONS

Module Name: Machine Learning for Scientific & Engineering Applications Module Code: UN-SAI-103 Module Type: Core		
Contact Hours/Week: 3L + 1T Module Credits: 04		
Module Objectives: Build a strong, practical ML workflow for scientific data, from data preparation and model selection to evaluation and interpretation. Students will understand ML as a data-driven baseline against which physics-informed models can later be evaluated.		
Units	Topics & Concepts	Project-Based Learning / Practical
Week 9	ML workflow; regression; linear and polynomial models; loss functions; regularization	Build a regression model to predict a scientific quantity. Compare a simple analytical baseline with ML.
Week 10	Classification; logistic regression; decision trees; Random Forest; feature engineering	Develop a classification pipeline using an engineering or scientific dataset.
Week 11	Bias-variance trade-off; overfitting; cross-validation; hyperparameter tuning; evaluation metrics	Conduct controlled model comparisons. Analyze generalization, data leakage and model errors.

Week 12	Clustering; PCA; feature scaling; model interpretation; scientific dataset challenges	Module Project: Complete a scientific ML pipeline with data cleaning, training, evaluation, error analysis and GitHub documentation.
Course Outcomes - CO1: Build end-to-end ML pipelines for scientific datasets. - CO2: Select and evaluate ML models using appropriate metrics and validation. - CO3: Establish data-driven baselines for future SciML/PINN comparisons.		
Books & References - An Introduction to Statistical Learning — James et al. - Hands-On Machine Learning — Aurélien Géron - Machine Learning for Engineering and Science Applications — NPTEL course		

MODULE 4: DEEP LEARNING, PYTORCH & DIFFERENTIABLE COMPUTING

Module Name: Deep Learning, PyTorch & Differentiable Computing Module Code: UN-SAI-104 Module Type: Core		
Contact Hours/Week: 3L + 1T Module Credits: 04		
Module Objectives: Train students to implement, debug, and evaluate neural networks using PyTorch. Establish the automatic differentiation and computational graph knowledge required for PDE residual computation and PINN development.		
Units	Topics & Concepts	Project-Based Learning / Practical
Week 13	Neural networks; perceptrons; activation functions; forward propagation; computational graphs	Implement a small neural network from scratch using NumPy and compare with PyTorch.
Week 14	Backpropagation; autograd; loss functions; SGD, Adam; learning-rate schedules	Build a PyTorch training pipeline. Inspect gradients,

		diagnose training problems and compare optimizers.
Week 15	CNNs; convolution and pooling; spatial data; introduction to sequence models	Train a CNN or sequence model on a suitable scientific image/time-series dataset.
Week 16	Automatic differentiation; higher-order derivatives; tensor operations; GPU workflows; reproducibility	Module Project: Train a neural network to approximate a known analytical function and compute its derivatives using PyTorch autograd.
Course Outcomes 1. CO1: Implement and train neural networks using PyTorch. 2. CO2: Explain backpropagation and automatic differentiation. 3. CO3: Compute model derivatives and prepare neural architectures for PINNs.		
Books & References 1. Deep Learning — Goodfellow, Bengio & Courville 2. Automatic Differentiation in Machine Learning: a Survey — Baydin et al. 3. Dive into Deep Learning — Zhang et al.		

MODULE 5: SCIENTIFIC MACHINE LEARNING & PINNs

Module Name: Scientific Machine Learning & Physics-Informed Neural Networks Module Code: UN-SAI-105 Module Type: Core		
Contact Hours/Week: 3L + 1T Module Credits: 04		
Module Objectives: Develop practical expertise in physics-informed and hybrid AI modelling. Students will formulate differential equations as learning constraints, implement PINNs using automatic differentiation, and evaluate their accuracy against numerical or analytical reference solutions.		
Units	Topics & Concepts	Project-Based Learning / Practical

Week 17	Scientific ML paradigm; data-driven vs physics-based models; hybrid modelling; inverse problems; PINN architecture	Formulate a scientific problem and identify its governing equations, data requirements, unknowns and validation strategy.
Week 18	PINN mathematical formulation; PDE residuals; initial/boundary conditions; collocation points; composite loss functions	Implement a basic PINN loss in PyTorch. Visualize residuals and examine the contribution of each loss term.
Week 19	PINNs for ODEs and forward problems; automatic differentiation; training and convergence challenges	Guided Project: Solve a 1D heat equation using a PINN. Compare predictions against an analytical or numerical solution.
Week 20	PINNs for nonlinear PDEs; Burgers equation; loss balancing; sampling strategies; error analysis	Implement a PINN for the 1D Burgers equation. Investigate the effects of collocation points and loss weights..
Week 21	Inverse PINNs; parameter estimation; noisy observations; scientific validation; introduction to neural operators	Solve a forward or inverse scientific problem using a PINN. Benchmark against a classical numerical solver and submit a technical report.

Course Outcomes

1. Formulate differential equations and physical constraints as PINN training objectives.
2. CO2: Implement forward and introductory inverse PINN models using PyTorch.
3. CO3: Validate PINN predictions using reference solutions, residuals and error metrics.

4. CO4: Diagnose common PINN training challenges and document experimental findings.
Books & References 1. Physics-Informed Neural Networks — Raissi, Perdikaris & Karniadakis (2019) 2. Scientific Machine Learning — Karniadakis et al. (2021), review paper 3. Numerical Methods for Engineers — Chapra & Canale 4. Understanding and Mitigating Gradient Flow Pathologies in PINNs — Wang, Teng & Perdikaris (2021)

MODULE 6: ADVANCED SCIENTIFIC AI, RESEARCH ENGINEERING & INDUSTRY CAPSTONE

Module Name: Advanced Scientific AI, Research Engineering & Industry Capstone Module Code: UN-SAI-106 Module Type: Capstone / Project		
Contact Hours/Week: 3L + 1T Module Credits: 04		
Module Objectives: Enable students to independently scope, develop, validate and communicate a scientific AI solution. Integrate PINNs/SciML, numerical benchmarks, reproducible experimentation, professional software practices and technical project defence into an industry/research-style deliverable.		
Units	Topics & Concepts	Project-Based Learning / Practical
Week 22	Advanced PINN training; adaptive sampling; loss balancing; optimization strategies; model limitations	Capstone Proposal: Select a scientific/engineering problem, review relevant literature, define objectives, baselines, metrics and project milestones.
Week 23	Neural operators; DeepONet/FNO concepts; hybrid physics-ML; model selection and computational trade-offs	Build a proof-of-concept for the chosen project. Select a suitable architecture and establish a baseline.

Week 24	Research computing; experiment tracking; Git branching; modular code; configuration management; reproducibility	Develop the main model, maintain a structured repository, track experiments and record design decisions.
Week 25	Benchmarking; ablation studies; uncertainty/error analysis; computational cost; technical reporting	Complete implementation and validation. Compare against numerical/ML baselines; prepare figures, tables and reproducible results.
Week 26	Technical communication; project defence; portfolio development; industry/research presentation	Final Capstone Evaluation: Demonstration, GitHub repository, technical report, presentation, viva and project defence.

Course Outcomes

1. CO1: Design and implement a scientific AI capstone addressing a defined problem.
2. CO2: Benchmark and validate models against numerical/data-driven baselines.
3. CO3: Apply reproducible research-engineering practices and professional Git/GitHub workflows.
4. CO4: Communicate and defend technical decisions, results, limitations and future improvements.

Books & References

1. Physics-Informed Neural Networks — Raissi, Perdikaris & Karniadakis (2019)
2. Fourier Neural Operator for Parametric Partial Differential Equations — Li et al. (2021)
3. An Introduction to Statistical Learning — James et al.

ASSESSMENT STRUCTURE

Component	Weight
Weekly assignments	15%
Coding/practical work	15%

Component	Weight
Module projects	20%
Quizzes/theory	10%
Final capstone	25%
Technical presentation/defence	10%
GitHub/portfolio/professionalism	5%
Total	100%

INSTRUCTOR BIO



Dishant Gupta is a B. Tech graduate in Engineering Physics from NIT Hamirpur, with experience at the intersection of Physics, AI, Computational Science, and Emerging Technologies. He is a former ISRO–SAC intern, where he worked on the design and computational modelling of metalenses, and a former Field & Technology Intern at Baker Hughes. He has also worked on AI/ML, Scientific Machine Learning, Physics-Informed Neural Networks (PINNs), and computational physics.

Dishant was among the top-performing participants in the WISER BQP Project Challenges 2026, working on quantum-enhanced approaches to scientific computing and PDE modelling.

Through UNISOLE SKILL AI LABS, he focuses on helping students build strong foundations in mathematics, programming, AI/ML, scientific computing, and research-oriented problem solving.

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